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CANDIDATE NUMBER

SYDNEY GRAMMAR SCHOOL



2018 Half-Yearly Examination

FORM VI

MATHEMATICS EXTENSION 2

Thursday 22nd February 2018

General Instructions

- Writing time — 1 hour 30 minutes
- Write using black pen.
- Board-approved calculators and templates may be used.

Total — 55 Marks

- All questions may be attempted.

Section I — 7 Marks

- Questions 1–7 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II — 48 Marks

- Questions 8–11 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your candidate number on each answer booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Write your candidate number on this question paper and hand it in with your answers.
- Place everything inside the answer booklet for Question Eight.

Checklist

- SGS booklets — 4 per boy
- Multiple choice answer sheet
- Reference Sheet
- Candidature — 74 boys

Examiner

LYL

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

Which expression is equivalent to $(1 - i)^3$?

- (A) $2 - 2i$
- (B) $-2 - 2i$
- (C) $2 + 2i$
- (D) $-2 + 2i$

QUESTION TWO

Given the complex numbers $z = 3i$ and $w = 1 - i\sqrt{5}$, what is the value of $\frac{z}{w}$?

- (A) $-\frac{\sqrt{5}}{2} + \frac{i}{2}$
- (B) $-\frac{3\sqrt{5}}{2} + \frac{3i}{2}$
- (C) $-\frac{\sqrt{5}}{6} + \frac{i}{6}$
- (D) $-\frac{\sqrt{5}}{2} - \frac{i}{2}$

QUESTION THREE

What is $-\sqrt{3} + i$ expressed in modulus-argument form?

- (A) $\sqrt{2} \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$
- (B) $2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right)$
- (C) $\sqrt{2} \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$
- (D) $2 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right)$

QUESTION FOUR

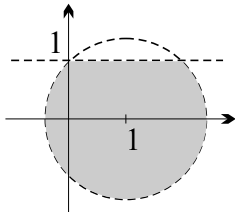
What is the general solution for the equation $\sin 2x = \sin \frac{\pi}{4}$?

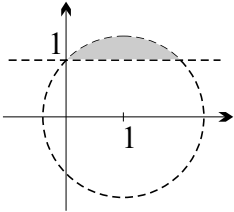
- (A) $x = \frac{\pi}{4} + 2n\pi$ or $x = \frac{3\pi}{4} + 2n\pi$ where n is an integer.
- (B) $x = \frac{\pi}{8} + n\pi$ or $x = -\frac{\pi}{8} + n\pi$ where n is an integer.
- (C) $x = \frac{\pi}{8} + n\pi$ or $x = \frac{3\pi}{8} + n\pi$ where n is an integer.
- (D) $x = \frac{\pi}{2} + 4n\pi$ or $x = \frac{3\pi}{2} + 4n\pi$ where n is an integer.

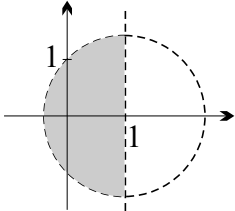
QUESTION FIVE

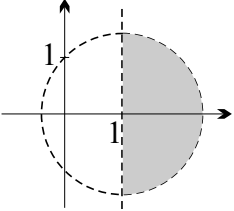
Which Argand diagram best represents the locus of z such that

$$|z - 1| \leq \sqrt{2} \text{ and } \operatorname{Im}(z + 4 + 3i) > 4?$$

- (A) 

(B) 

(C) 

(D) 

QUESTION SIX

Let $z = x + iy$ and $z, \bar{z}, -z, i\bar{z}$ and iz be represented by the points P, Q, R, S and T on the Argand diagram. Which geometric statement is incorrect?

- (A) The points Q and P are reflections of each other in the real axis.
- (B) The points R and P are rotations of each other by π about the origin.
- (C) The point S is a reflection of P in the line $y = -x$.
- (D) The point T is the result of rotating P by $\frac{\pi}{2}$ anticlockwise about the origin.

QUESTION SEVEN

Which integral is equivalent to $\int \sqrt{\frac{x+1}{x+3}} dx$?

(A) $\int \frac{\sqrt{x^2-4x+3}}{x+3} dx$

(B) $\int \sqrt{\frac{x^2+2x-3}{x^2-9}} dx$

(C) $\int \frac{x+1}{\sqrt{x^2-2x-1}} dx$

(D) $\int \frac{x+1}{\sqrt{x^2+4x+3}} dx$

_____ End of Section I _____

Examination continues next page ...

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION EIGHT (12 marks) Use a separate writing booklet.

Marks

(a) Let $z = 4 - 5i$ and $w = -4 + i$. Find:

(i) $|z - w|$

1

(ii) $\arg(z + w)$

1

(b) Let $z = x + iy$. Prove that:

(i) $\operatorname{Re}(z + \bar{z}) = 2x$

1

(ii) $z\bar{z} = |z|^2$

1

(c) Differentiate $x^2 - y^3 + xy^2 = 3$ with respect to x and hence find $\frac{dy}{dx}$.

2

(d) Find $\int \frac{6x + 4}{x^2 + 16} dx$.

2

(e) (i) Find the two square roots of $-21 - 20i$.

2

(ii) Hence, or otherwise, solve $z^2 - (4 + i)z + 9 + 7i = 0$.

2

QUESTION NINE (12 marks) Use a separate writing booklet.**Marks**

(a) Determine $\int \sin^4 x \cos^3 x \, dx$. **2**

(b) Use mathematical induction to prove that **3**

$$n! \geq 2^{n-1}$$

for all positive integers n .

[NOTE: by definition $n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$]

(c) (i) Find real numbers A , B and C such that **2**

$$\frac{9x^2}{(x-1)^2(x+2)} \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}.$$

(ii) Hence find $\int \frac{9x^2}{(x-1)^2(x+2)} \, dx$. **1**

(d) Using integration by parts, find $\int x e^{2x} \, dx$. **2**

(e) In an Argand diagram, the points P and Q represent the complex numbers w and $\frac{1}{w}$ respectively. The interval PQ is produced to R such that $QR = 2PQ$. Find the complex number represented by the point R . Leave your answer in terms of w . **2**

QUESTION TEN (12 marks) Use a separate writing booklet.

Marks

(a) Find $\int \frac{1}{x(\log_e x)^3} dx$. 1

(b) (i) Show that the derivative of $\sec \theta$ is $\sec \theta \tan \theta$. 1

(ii) Use the substitution $x = \sec \theta - 1$ to find the exact value of 3

$$\int_{\sqrt{2}-1}^1 \frac{2}{(x+1)\sqrt{x^2+2x}} dx.$$

(c) Let $I_n = \int_1^e \frac{(\ln x)^n}{x^2} dx$ for $n \geq 0$.

(i) Show that $I_n = -\frac{1}{e} + nI_{n-1}$ for $n \geq 1$. 2

(ii) Hence find I_2 . 2

(d) The complex number z satisfies $|z - 3i| \leq 4$.

(i) Find the least possible value of $|z - 5 + 2i|$. 2

(ii) For what value of z is the least possible value attained? 1

QUESTION ELEVEN (12 marks) Use a separate writing booklet.

Marks

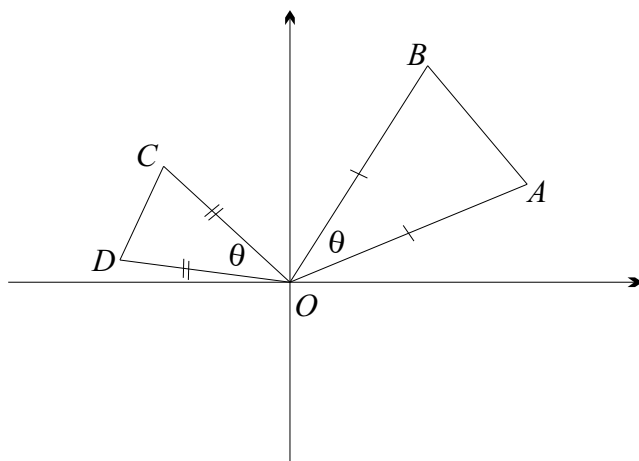
- (a) Sketch the locus of the points such that $|z^2 - \bar{z}^2| < 4$.

2

- (b) Use the substitution $t = \tan \frac{x}{2}$ to determine $\int \frac{1}{4 + 3 \sin x} dx$.

3

(c)



The Argand diagram above shows the points O , A , B , C and D . These points represent the complex numbers, 0 , a , b , c and d respectively. Triangles OAB and OCD are similar and isosceles with $\angle AOB = \angle COD = \theta$ where θ is acute.

Let $\omega = \cos \theta + i \sin \theta$.

- (i) Explain why $d = \omega c$.

1

- (ii) Find the complex number a in terms of b .

1

- (iii) Using complex numbers, show that the lengths AC and BD are equal.

2

- (d) Find $\int x e^x \sin x dx$.

3

_____ End of Section II _____

END OF EXAMINATION

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- Record your multiple choice answers by filling in the circle corresponding to your choice for each question.
- Fill in the circle completely.
- Each question has only one correct answer.

Question One

A ☐ B ☐ C ☐ D ☐

Question Two

A ☐ B ☐ C ☐ D ☐

Question Three

A ☐ B ☐ C ☐ D ☐

Question Four

A ☐ B ☐ C ☐ D ☐

Question Five

A ☐ B ☐ C ☐ D ☐

Question Six

A ☐ B ☐ C ☐ D ☐

Question Seven

A ☐ B ☐ C ☐ D ☐

HALF-YEARLY EXTENSION 2, 2018

MC

Q1 $(1-i)^3$

let $z = 1-i$

$$z^2 = (1-i)^2$$

$$= -2i$$

$$z^3 = z^2 \times z$$

$$= -2i(1-i)$$

$$= -2i - 2$$

(B)

Q2

$$\frac{3i}{1-i\sqrt{5}}$$

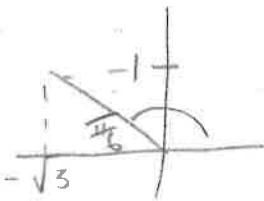
$$\times \frac{1+i\sqrt{5}}{1+i\sqrt{5}}$$

$$= \frac{3i - 3\sqrt{5}}{6}$$

$$= \frac{i}{2} - \frac{\sqrt{5}}{2}$$

(A)

Q3 $-\sqrt{3} + i = 2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$



$$\theta = \frac{5\pi}{6}$$

$$r = 2$$

(D)

Q4

$$\sin 2x = \sin \frac{\pi}{4}$$

$$2x = \frac{\pi}{4} + 2n\pi$$

$$x = \frac{\pi}{8} + n\pi$$

$$\text{or } 2x = \left(\pi - \frac{\pi}{4}\right) + 2n\pi$$

$$2x = \frac{3\pi}{4} + 2n\pi$$

(C)

$$x = \frac{3\pi}{8} + n\pi$$

MC

Q 5

$$|z - 1| \leq \sqrt{2}$$



region inside a circle centre (1,0)
radius $\sqrt{2}$
solid boundary

$$\operatorname{Im}(z + 4 + 3i) > 4$$

$$z + 4 + 3i = z - (\bar{4} + \bar{3}i)$$

$$\text{Let } z = x + iy$$

$$x + iy + 4 + 3i = x + 4 + i(y + 3)$$

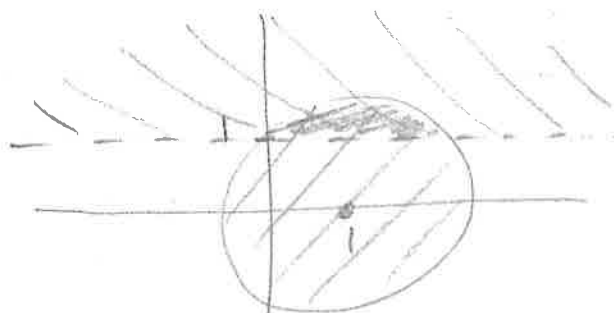
$$\operatorname{Im}(z + 4 + 3i) = y + 3$$

$$y + 3 > 4$$

$$y > 1$$

Dotted horizontal line
shaded above the line

$$y = 1$$



(B)

$$|z - 1| \leq \sqrt{2} \quad \text{and} \quad \operatorname{Im}(z + 4 + 3i) > 4$$

MC/

Q6

The point S is a reflection of P
in line $y = x$. NOT $y = -x$

(C)

Q7

$$\int \frac{\sqrt{x+1}}{\sqrt{x+3}} \times \frac{\sqrt{x+1}}{\sqrt{x+1}} dx$$

$$= \int \frac{x+1}{\sqrt{(x+3)(x+1)}} dx$$

$$= \int \frac{x+1}{\sqrt{x^2+4x+3}} dx$$

(D)

Section II

Q8 $z = 4 - 5i$ $w = -4 + i$

a) i) $z - w = 4 - 5i - (-4 + i)$

$$= 8 - 6i$$

$$|z - w|^2 = 8^2 + (-6)^2$$

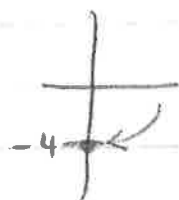
$$= 100$$

$$|z - w| = 10 \quad \checkmark$$

ii) $z + w = 4 - 5i + (-4 + i)$

$$= -4i$$

$$\arg(z + w) = -\frac{\pi}{2} \quad \checkmark$$



b) i) Since $z = x + iy$ $\bar{z} = x - iy$

$$z + \bar{z} = x + iy + x - iy$$

$$= 2x$$

$$\operatorname{Re}(z + \bar{z}) = 2x \quad \text{as} \quad \operatorname{Im}(z + \bar{z}) = 0$$

ii) $z\bar{z} = (x + iy)(x - iy)$

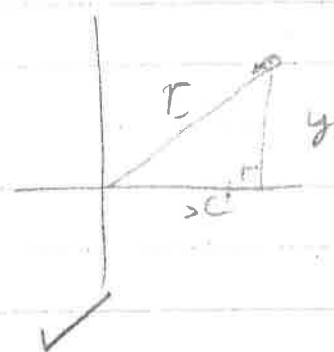
$$= x^2 - i^2 y^2$$

$$= x^2 + y^2$$

$$|z|^2 = r^2$$

$$= x^2 + y^2$$

$$= z\bar{z}$$



c) $x^2 - y^2 + 2xy = 3$

$$2x - 3y^2 \frac{dy}{dx} + 2y \frac{dy}{dx} + y^2 = 0$$

must have differentiated 3 terms correctly

$$\frac{dy}{dx} (2xy - 3y^2) = -2x - y^2$$

$$\frac{dy}{dx} = \frac{-(2x + y^2)}{2xy - 3y^2}$$

$$Q8 \text{ c) } \int \frac{6x+4}{x^2+16} dx$$

$$= \int \frac{6x}{x^2+16} dx + \int \frac{4}{x^2+16} dx$$

$$= 3 \int \frac{2x}{x^2+16} dx + \int \frac{4}{x^2+16} dx \quad \checkmark$$

$$= 3 \ln(x^2+16) + 4 \cdot \frac{1}{4} \tan^{-1} \frac{x}{4} + C$$

$$= 3 \ln(x^2+16) + \tan^{-1} \frac{x}{4} + C \quad \checkmark$$

$$d) 1) \text{ Let } \lambda = x+iy$$

$$\text{Let } (x+iy)^2 = -21-20i$$

$$x^2 - y^2 = -21$$

$$2xy = -20$$

$$xy = -10$$

$$\left. \begin{array}{l} x^2 - y^2 = -21 \\ 2xy = -20 \\ xy = -10 \end{array} \right\} \quad \checkmark$$

Different signs

$$x=2 \quad y=-5 \quad \text{or} \quad x=-2 \quad y=5$$

$$\lambda = 2-5i \quad \text{or} \quad \lambda = -(2-5i) \quad \checkmark$$

$$u) \quad z^2 - (4+i)z + (9+7i) = 0$$

$$a = 1$$

$$b = -(4+i)$$

$$c = 9+7i$$

$$\begin{aligned} \Delta &= (4+i)^2 - 4 \times 1 \times (9+7i) \\ &= 16 + 8i + i^2 - 36 - 28i \\ &= 16 - 1 - 36 - 20i \\ &= -21 - 20i \quad \checkmark \end{aligned}$$

$$z = \frac{4+i + 2-5i}{2}$$

$$= \frac{6-4i}{2}$$

$$z = 3-2i$$

$$\text{or} \quad z = \frac{4+i - (2-5i)}{2}$$

$$= \frac{2+6i}{2}$$

$$z = 1+3i \quad \checkmark$$

Q9 a) $\int \sin^4 x \cos^3 x dx$

$$= \int \sin^4 x (1 - \sin^2 x) \cos x dx$$

$$= \int (\sin^4 x - \sin^6 x) \cos x dx \quad \checkmark \quad \text{let } u = \sin x$$

$$= \int u^4 - u^6 du \quad \frac{du}{dx} = \cos x$$

$$= \frac{u^5}{5} - \frac{u^7}{7} + C \quad du = \cos x dx$$

$$= \frac{\sin^5 x}{5} - \frac{\sin^7 x}{7} + C \quad \checkmark$$

b) Prove by MI $n! \geq 2^{n-1}$ for positive integer n

Part A When $n=1$ LHS = $1!$ RHS = 2^{1-1}

$$= 1 \quad = 2^0 = 1 \quad \checkmark$$

$$\text{LHS} \geq \text{RHS}$$

Statement is true for $n=1$

Part B Assume statement is true for $n=k$ where k is a positive integer

$$\text{i.e. } k! \geq 2^{k-1} \quad *$$

We must prove statement true for $n=k+1$

$$\text{i.e. } (k+1)! \geq 2^k$$

Beginning with the induction hypothesis *

$$k! \geq 2^{k-1}$$

$$(k+1) \times k! \geq (k+1) 2^{k-1}$$

$$\geq 2 \times 2^{k-1} \quad \text{since } k \geq 1 \quad \checkmark$$

$$= 2^k$$

Part C It follows from Parts A & B by MI the statement holds true for all positive integers n .

$$09 c) \int \frac{9x^2}{(x-1)^2(x+2)} dx \equiv \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

$$9x^2 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

$$\text{if } x=1 \quad \begin{array}{l} 9 = 3B \\ \underline{B = 3} \end{array} \quad \checkmark$$

$$\text{if } x = -2 \quad 36 = (-3)^2 C$$

$$\text{coefficient of } \underline{\frac{C}{x^2}} = 4$$

$$A + C = 9$$

$$A + 4 = 9$$

$$\underline{A = 5} \quad \checkmark$$

$$\int \frac{9x^2}{(x-1)^2(x+2)} dx = \int \frac{5}{x-1} + \frac{3}{(x-1)^2} + \frac{4}{x+2} dx \quad \checkmark$$

$$= 5 \ln(x-1) - \frac{3}{x-1} + 4 \ln(x+2) + C$$

$$= \ln(x-1)^5 + \ln(x+2)^4 - \frac{3}{x-1} + C$$

$$= \ln[(x-1)^5(x+2)^4] - \frac{3}{x-1} + C$$

09 d) $I = \int x e^{2x} dx$

let $u = x$
 $u' = 1$

$v' = e^{2x}$
 $v = \frac{e^{2x}}{2}$

$$I = \frac{x e^{2x}}{2} - \int \frac{e^{2x}}{2} dx \quad \checkmark$$

$$= \frac{x e^{2x}}{2} - \frac{e^{2x}}{4} + C \quad \checkmark$$

e) $\overrightarrow{PQ} = \frac{1}{w} - w$

Diagram is
 if P is in
 Quadrant 1

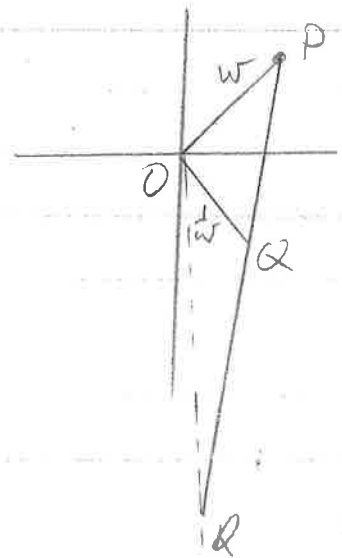
$$|QR| = 2|PQ|$$

$$\begin{aligned} \overrightarrow{PR} &= 3\overrightarrow{PQ} \\ &= 3\left(\frac{1}{w} - w\right) \end{aligned}$$

$$= \frac{3}{w} - 3w \quad \checkmark$$

$$\begin{aligned} \vec{OR} &= \vec{OP} + \vec{PR} \\ &= w + \frac{3}{w} - 3w \end{aligned}$$

$$= \frac{3}{w} - 2w \quad \checkmark$$



Q10 a) $\int x \left(\frac{1}{\log_e x} \right)^3 dx$

$$= \int \frac{1}{x} \times (\log_e x)^{-3} dx$$

$$= \int u^{-3} du$$

$$= \frac{u^{-2}}{-2} + C$$

$$= -\frac{1}{2(\log_e x)^2} + C$$

Let $u = \log_e x$

$$\frac{du}{dx} = \frac{1}{x}$$

$$du = \frac{dx}{x}$$

b) i) $\frac{d}{d\theta} \sec \theta = \frac{d}{d\theta} \left(\frac{1}{\cos \theta} \right)$

$$= \frac{d}{d\theta} (\cos \theta)^{-1}$$

$$= -(\cos \theta)^{-2} \times -\sin \theta$$

$$= \frac{\sin \theta}{\cos^2 \theta}$$

$$= \frac{1}{\cos \theta} \times \frac{\sin \theta}{\cos \theta}$$

$$= \sec \theta \tan \theta$$

ii) next page

Q10 b) ii) $x = \sec \theta - 1$
 $\frac{dx}{d\theta} = \sec \theta + \tan \theta$

$$\int_{\sqrt{2}-1}^1 \frac{2}{(x+1)\sqrt{x^2+2x}} dx$$

$$\begin{aligned} x &= \sqrt{2}-1 \\ \sqrt{2}-1 &= \sec \theta - 1 \\ \sqrt{2} &= \sec \theta \\ \cos \theta &= \frac{1}{\sqrt{2}} \end{aligned}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{2 \sec \theta + \tan \theta d\theta}{(\sec \theta - 1 + 1)(\sqrt{\sec^2 \theta - 1 + 2(\sec \theta - 1)})}$$

$\theta = \frac{\pi}{4}$

award 1 mark if limits incorrect

$$\begin{aligned} x &= 1 \\ \sec \theta &= 2 \\ \cos \theta &= \frac{1}{2} \end{aligned}$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{2 \sec \theta + \tan \theta d\theta}{\sec \theta \sqrt{\sec^2 \theta - 2 \sec \theta + 1 + 2 \sec \theta - 2}}$$

$\theta = \frac{\pi}{3}$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{2 + \tan \theta d\theta}{\sqrt{\tan^2 \theta}}$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

must have correct limits.

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 2 d\theta$$

$$= 2 \left[\theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$= 2 \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$$

$$= 2 \left(\frac{4\pi - 3\pi}{12} \right)$$

$$= \frac{\pi}{6}$$

$$\text{Q10 c)} \quad I_n = \int_1^e \frac{(\ln x)^n}{x^2} dx \quad n \geq 0$$

By integration by parts

$$u = (\ln x)^n \quad v' = x^{-2}$$

$$u' = n(\ln x)^{n-1} \times \frac{1}{x}$$

$$= \frac{n(\ln x)^{n-1}}{x}$$

$$v = \frac{x^{-1}}{-1}$$

$$= -\frac{1}{x}$$

$$I_n = \left[-\frac{1}{x} \times (\ln x)^n \right]_1^e - \int_1^e -\frac{n(\ln x)^{n-1}}{x^2} dx$$

$$= \left[\frac{1}{e} (\ln e)^n - 1 \times (\ln 1)^n \right] + n \int_1^e \frac{(\ln x)^{n-1}}{x^2} dx$$

$$= -\frac{1}{e} + 0 + n I_{n-1} \quad \text{for } n \geq 1$$

$$I_2 = -\frac{1}{e} + 2 I_1$$

$$I_1 = -\frac{1}{e} + I_0$$

$$I_0 = \int_1^e \frac{1}{x^2} dx$$

$$= -\left[\frac{1}{x} \right]_1^e$$

$$= -\left(\frac{1}{e} - 1 \right)$$

$$= 1 - \frac{1}{e}$$

$$I_1 = -\frac{1}{e} + \left(1 - \frac{1}{e} \right)$$

$$= 1 - \frac{2}{e}$$

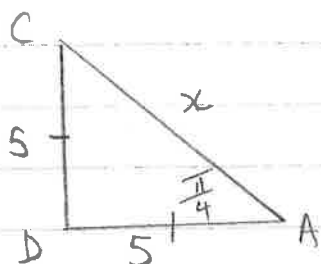
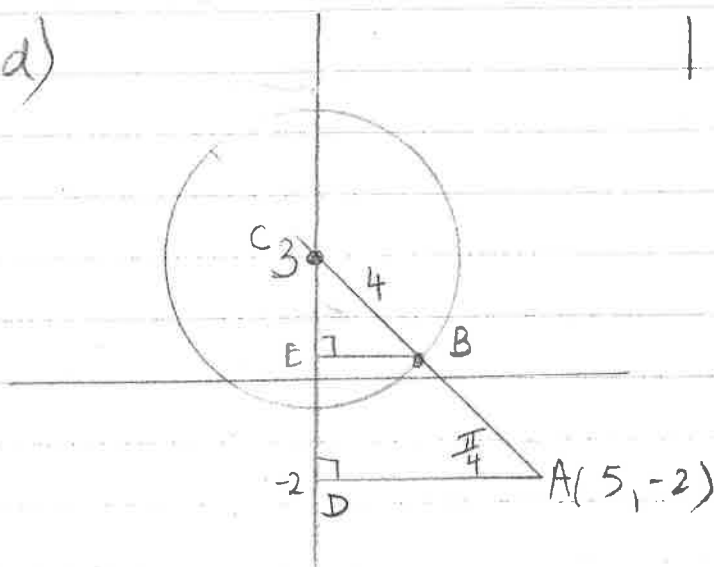
$$I_2 = -\frac{1}{e} + 2 \left(1 - \frac{2}{e} \right)$$

$$= 2 - \frac{5}{e} \quad \checkmark$$

Q10 d)

$$|z - 3i| \leq 4$$

circle centre $3i$
radius 4



$$\begin{aligned} x^2 &= 5^2 + 5^2 \\ &= 50 \\ x &= 5\sqrt{2} \end{aligned}$$

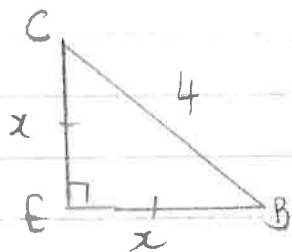
✓

$$AC = 5\sqrt{2}$$

$$BC = \text{radius of circle} = 4$$

$$AB = AC - BC = 5\sqrt{2} - 4$$

✓



Right angled isosceles

$$x^2 + x^2 = 4^2$$

$$2x^2 = 16$$

$$x^2 = 8$$

$$x = 2\sqrt{2}$$

$$y = 3 - 2\sqrt{2}$$

$$B(2\sqrt{2}, 3 - 2\sqrt{2})$$

✓

The complex number z is $2\sqrt{2} + (3 - 2\sqrt{2})i$

Q11 a)

$$|z^2 - (\bar{z})^2| \leq 4$$

$$\text{Let } z = x + iy$$

$$\text{LHS} = |(x+iy)^2 - (x-iy)^2|$$

$$= |[x+iy - (x-iy)][x+iy + x-iy]|$$

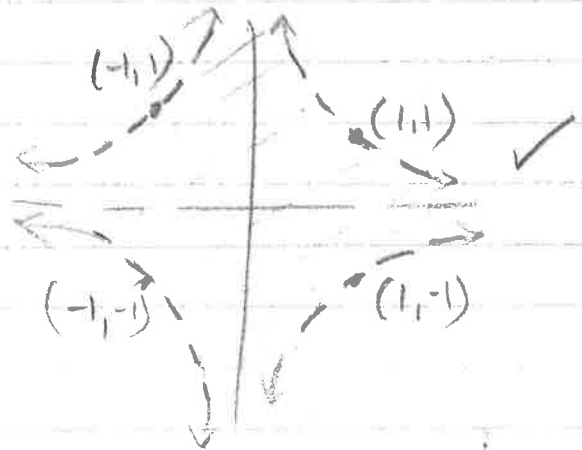
$$= |(2iy)(2x)|$$

$$= |4ixy|$$

$$4|xy| < 4$$

$$|xy| < 1$$

$$-1 < xy < 1$$



Q11

$$b) \int \frac{1}{4+3\sin x} dx$$

$$= \int \frac{\frac{2dt}{1+t^2}}{\frac{4(1+t^2)}{1+t^2} + \frac{3(2t)}{1+t^2}} \quad \checkmark$$

$$= \int \frac{2}{1+t^2} \times \frac{1+t^2}{4+4t^2+6t} dt$$

$$= \int \frac{2}{4t^2+6t+4} dt$$

$$= \int \frac{2}{4(t^2 + \frac{3}{2}t + 1)} dt$$

$$= \frac{1}{2} \int \frac{dt}{(t^2 + \frac{3}{2}t + (\frac{3}{4})^2) - \frac{9}{16} + 1}$$

$$= \frac{1}{2} \int \frac{dt}{(t + \frac{3}{4})^2 + \frac{7}{16}} \quad \checkmark$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{\frac{7}{16}}} \tan^{-1} \left(\frac{t + \frac{3}{4}}{\frac{\sqrt{7}}{4}} \right) + C$$

$$= \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{4t+3}{\sqrt{7}} \right) + C$$

$$= \frac{2}{\sqrt{7}} \tan^{-1} \left(\frac{4 + \tan \frac{x}{2} + 3}{\sqrt{7}} \right) + C \quad \checkmark \quad \begin{array}{l} \text{must} \\ \text{back} \\ \text{substitute} \end{array}$$

$$t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \frac{1}{2} (1 + \tan^2 \frac{x}{2})$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$a^2 = \frac{7}{16}$$

$$a = \frac{\sqrt{7}}{4}$$

Q11
i) $\vec{OD}$ is $\vec{OC}$ rotated anticlockwise about the origin by θ

$$\begin{aligned} d &= c \times e^{i\theta} \\ &= c \times w \\ &= wc \end{aligned}$$

ii) $\vec{OA}$ is $\vec{OB}$ rotated clockwise about the origin by θ ($= -\theta$)

$$\begin{aligned} a &= [\cos(-\theta) + i\sin(-\theta)] \times b \\ &= (\cos \theta - i\sin \theta) b \\ &= \bar{w} b \end{aligned}$$

Also $b = \frac{a}{\bar{w}} = wa$

iii) $\vec{AC} = \vec{AO} + \vec{OC}$
 $= -a + c = c - a$

$$\begin{aligned} w\vec{AC} &= wc - wa \\ &= d - b \\ &= \vec{BD} \end{aligned}$$

$$\begin{aligned} |\vec{BD}| &= |w\vec{AC}| \\ &= |w| |\vec{AC}| \end{aligned}$$

$$= 1 \times |\vec{AC}| \quad \text{since } |w| = 1$$

$$\therefore |\vec{BD}| = |\vec{AC}| \text{ as required.}$$

$$I = \int x e^x \sin x \, dx$$

$$= \int x \sin x e^x \, dx$$

$$u = x \sin x \quad v' = e^x$$

$$u' = \sin x + x \cos x \quad v = e^x$$

$$I = x e^x \sin x - \int e^x \sin x + x e^x \cos x \, dx$$

$$= x e^x \sin x - \int e^x \sin x \, dx - \int x e^x \cos x \, dx$$

$$\text{Let } I_1 = \int e^x \sin x \, dx$$

$$u = \sin x$$

$$u' = \cos x$$

$$v' = e^x$$

$$v = e^x$$

$$I_1 = e^x \sin x - \int e^x \cos x \, dx$$

$$\text{Let } I_2 = \int e^x \cos x \, dx$$

$$u = \cos x$$

$$u' = -\sin x$$

$$v' = e^x$$

$$v = e^x$$

$$I_2 = e^x \cos x - \int -e^x \sin x \, dx$$

$$= e^x \cos x + I_1$$

$$I_1 = e^x \sin x - (e^x \cos x + I_1)$$

$$= e^x \sin x - e^x \cos x - I_1$$

$$2I_1 = e^x (\sin x - \cos x)$$

$$I_1 = \frac{e^x}{2} (\sin x - \cos x)$$



$$I_3 = \int x e^x \cos x \, dx$$

$$= \int x \cos x e^x \, dx$$

$$u = x \cos x$$

$$u' = \cos x + x(-\sin x)$$

$$v' = e^x$$

$$v = e^x$$

$$I_3 = x e^x \cos x - \int e^x \cos x - x e^x \sin x \, dx$$

$$= x e^x \cos x - I_2 + \int x e^x \sin x \, dx$$

$$= x e^x \cos x - e^x \cos x - I_1 + I$$

$$I = x e^x \sin x - I_1 - I_3$$

$$= x e^x \sin x - \cancel{I_1} - (x e^x \cos x - e^x \cos x - \cancel{I_1} + I)$$

$$I = x e^x \sin x - x e^x \cos x + e^x \cos x - I$$

$$2I = e^x (x \sin x - x \cos x + \cos x)$$

$$I = \frac{e^x}{2} (x \sin x - x \cos x + \cos x) + C$$